BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 303 · Appendix 5d OCR notation challenge

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why can OCR array[5] access index4 but not index5?

Q2. What is byRef?

Q3. What does readLine return?

Q4. In C# array [2,3,5,7], what is a[2] and total sum?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

array — Fixed-length zero-indexed sequence.

byVal — Pass a value/copy.

byRef — Pass a reference to original.

readLine — Read the next file line.

New learning

Use exact indexing — OCR arrays use zero-based indexing; the number in array declaration is the number of elements.

Trace subroutines and parameters — A procedure does not produce a return value; a function returns a result. byVal passes a copy; byRef allows the original parameter to change.

Read file operations — openRead, readLine, endOfFile and close follow the OCR Appendix 5d conventions. Always close an opened file.

Worked example

Trace a 3-element OCR array and identify the valid indexes.

1. Predict first — Array declaration creates three positions: indexes 0,1,2.

2. Check reasoning — scores[1] holds 4, so output is 4.

We Do · Practise with guidance

Trace a function and a procedure that changes an argument by reference.

```
procedure addOne(x:byRef)
 x=x+1
endprocedure
n=6
addOne(n)
print(n)
```

Predict first

The parameter is passed by reference, so assignment to x changes the caller's n.

Check reasoning

After addOne(n), n=7 and print displays 7.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace the OCR reference-language array below. State the output and explain why accessing a[3] would be invalid. [5]
CODE:
array a[3]
a[0] = 4
a[1] = 8
a[2] = 12
print(a[2])

Question 2

Write OCR exam reference language to read and print every line of the file scores.txt, and then close it. [7]

Question 3

Compare procedure addTwo(x:byVal) with addTwo(x:byRef), each executing x=x+2, for caller n=5. [6]

Exit Ticket · Answer before checking

array values[4] can validly access values[4].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Five items are indexed 0–4.

Q2. Parameter passing allowing modification of caller's underlying variable.

Q3. Next line of text from a file.

Q4. a[2]=5; total=17.

You Do pages · Independent questions

Q1. Outputs 12 [2]; valid indexes are 0, 1 and 2 [2]; a[3] is out of range [1].

Q2. Marking: openRead with the correct filename [2]; loop until endOfFile() [2]; readLine() and print each line [2]; close the loop and file [1].
CODE:

```
f = openRead("scores.txt")
while NOT f.endOfFile()
print(f.readLine())
endwhile
f.close()
```

Q3. byVal called with n=5 uses copy and n remains5 [3]; byRef updates n to7 [3].

Exit Ticket · Model answer

False. Valid positions are 0–3.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.